SatQuery AI — Execution Report

SIH26167 | ISRO / Space Technology
Problem Statement ID: SIH26167 | Run: run_1788869629513_b32d64
Generated: 2026-09-08 12:13:49 UTC | Latency: 1474 ms

Query

What changed between these two dates, and where did the change occur?

Input Tiles

Tile ID	Sensor	GSD (m)	Bands	Timestamp	Clear
8f931eb903_r0_c0	s2_msi	10.0	blue, green, nir, red, swir1, swir2	2024-02-10	100.0%
b416cd5236_r0_c0	s2_msi	10.0	blue, green, nir, red, swir1, swir2	2024-11-21	81.4%

Execution Trace

The statement evaluates the observable execution trace. Every step below corresponds directly to one of the six controller duties named in the problem statement.

#	Stage	Step	Tool / Model	Confidence	Summary
0	0	ingest	stage0-ingest 0.1.0	--	open_validate: s2_t1_2024-02-10.tif: 12 band(s) 512x512 uint16, g
1	0	ingest	stage0-ingest 0.1.0	--	open_validate: s2_t2_2024-11-21.tif: 12 band(s) 512x512 uint16, g
2	1	interpret	router-trained v2	0.74	intent=change (p=0.740) via router-trained v2; target_class=None;
3	1	check_compatibility		--	5/5 fields passed; action=proceed
4	1	select_tool	satquery-change 0.1.0	--	satquery-change v0.1.0 selected by declarative filter (task=change
5	1	bind_params	satquery-change 0.1.0	--	bound ['conf_thresh', 'dead_band'] against declared bounds
6	2	specialist	satquery-change 0.1.0	0.76	conf=0.764; Bi-temporal pair (2024-02-10 -> 2024-11-21). Class ch
7	3	integrate	satquery-change 0.1.0	0.74	raw=0.764 -> calibrated=0.745; answered; 0 conflict(s)

Compatibility Gate Verdicts

Five-field compatibility check per section 4.2. Passes are recorded explicitly -- a silent pass is indistinguishable from a check that never ran.

Field	Passed	Detail
number	PASS	2 tile(s) supplied; change requires exactly 2 (a bi-temporal pair)
modality	PASS	modalities ['optical', 'optical'] accepted for change
format	PASS	sensor profile(s) resolved: ['s2_msi']; 2/2 georeferenced
metadata	PASS	GSD [10.0, 10.0] m; timestamps ['2024-02-10', '2024-11-21']; timestamps distinct
compatibility	PASS	footprint IoU 1.000 (threshold 0.9); residual alignment shift 0.14 px (tolerance 2.0 px) measured by phase correlat

Answer

[ANSWER] Confidence: 74% | Intent: change | Output kind: mask

Bi-temporal pair (2024-02-10 -> 2024-11-21). Class changes (dead-band 2pp): vegetation increased (71.6% -> 84.8%); built up decreased (15.4% -> 7.7%); burn scar decreased (11.5% -> 7.3%). 19.5% of the tile shows spectral change (511 ha).

Calibration

raw_confidence	0.7644
temperature	1.1
calibrated	0.7446
gate_degraded	False
gsd_fit	1.0
conflict_penalty	0
abstain_threshold	0.38
abstained	False
task	change

Reproducibility

Input hash + weights hash is what an ISRO / SAC reviewer will ask for. Both are stored on every trace step.

Step	input_hash	weights_hash
ingest	ccf261eedcd8c7cd	--
ingest	3e69c74629f2d82f	--
select_tool	--	uninitialised
specialist	ccf261eedcd8c7cd|3e69c74629f2d82f	3c7f3f85665710cb

Software


```
{ "satquery": "0.1.0", "torch": "2.5.1+cu121", "rasterio": "1.5.1", "numpy": "2.5.2",
"torchgeo": "0.10.0", "scipy": "1.18.0" }
```